

Raphaël Terrine

3rd year PhD student at Polytechnique Institute

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Research Interests

I am interested in the study of Partial Differential Equations, particularly in the time domain, both numerically and theoretically and the study of inverse problems using optimal control and one shot methods. I'm also interested in the implementation and the analysis of the Finite Elements Method.

Education

2023 – 2026 PhD in Applied Mathematics ENSTA Paris/Polytechnique

Thesis: Identification of bottom deformations of the ocean from surface measurements.

2020 – 2023 Engineering school ENSTA Paris

Specialisation : Applied maths and waves propagation.

2022 – 2023 Master's degree in applied Mathematics Polytechnique Institute

Specialisation : Modelisation, Simulation and Analysis.

2018 – 2020 French Preparatory Classes MPSI-MP* Lycée Henri IV

Two-year intensive preparatory program for competitive entrance to French Grandes Écoles.

Professional experiences

Summer 2023 Pre PhD internship in POEMS team ENSTA Paris

Identification of bottom deformations of the ocean from surface measurements.

Summer 2022 Research Internship supervised by G. Bal University of Chicago

Analysis of transport along the boundaries of domains generated by Coriolis forces in geophysical fluid flows.

Publications

2024 L. Bourgeois, J-F. Mercier, R. Terrine Inverse Problems and Imaging

Identification of bottom deformations of the ocean from surface measurements.

2026 L. Bourgeois, P. Moireau, R. Terrine

Numerical analysis of an optimal control approach to solve a tsunami inverse problem.

2026 L. Bourgeois, P. Moireau, R. Terrine

An inverse tsunami problem in the time domain: a well-posedness analysis of the forward problem and an inversion strategy based on a mixed formulation of the Tikhonov regularization.

Talks

28 August 2026 **Invited speaker at Wave-Ned 2026** Delft
Convergence of a numerical scheme for solving an inverse problem : Application to tsunami reconstruction.

29 October 2025 **Invited speaker at PICOF'2025** Hamammet
Convergence of a numerical scheme for solving an inverse problem : Application to tsunami reconstruction.

5 June 2025 **SMAI 2025** Carcans
Identification of bottom deformations from surface measurements.

4 July 2024 **Waves 2024** Berlin
About the identification of tsunamis from surface measurements.

Teaching

2023-2025 **APM-3MA02-TA : L3 level class** ENSTA Paris
Outils élémentaires d'analyse pour les equations aux dérivées partielles.

Technical Skills

Programming: Python, C++, MATLAB

Tools: LaTeX, Git

Languages: French (native), English (fluent)